Complex Numbers: Cartesian Form & Algebra

Section 1: Cartesian Arithmetic & Complex Conjugate

1 Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

a. $5 + 14i$

b. $-3 - 8i$

c. $5 - 14i$

d. $-3 + 14i$

2 Express $\frac{4+2i}{1-i}$ in Cartesian form $a + bi$.

a. $1 + 3i$

b. $3 + 3i$

c. $1 - 3i$

d. $2 + 2i$

3 If $z = x + iy$, what is $z \cdot z^*$ (where z^* is the complex conjugate)?

a. $x^2 + y^2$

b. $x^2 - y^2$

c. $x^2 + 2ixy - y^2$

d. $2x$

Section 2: Modulus and Argument

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4 Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

a. $|z| = 2, \text{Arg}(z) = \frac{5\pi}{6}$

b. $|z| = 2, \text{Arg}(z) = -\frac{\pi}{6}$

c. $|z| = 4, \text{Arg}(z) = \frac{2\pi}{3}$

d. $|z| = 2, \text{Arg}(z) = \frac{7\pi}{6}$

5 What is the modulus of $\frac{3+4i}{1-2i}$?

a. $\sqrt{5}$

b. $\frac{5}{3}$

c. 5

d. $\frac{25}{5} = 5$

6 Find the principal argument of $z = -5i$.

a. $-\frac{\pi}{2}$

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{2}$

d. π

Section 3: Solving Complex Equations

7 Solve the equation $2z + 3z^* = 15 - 2i$ for $z = x + iy$.

a. $z = 3 + 2i$

b. $z = 3 - 2i$

c. $z = 5 - 2i$

d. $z = 2 + 3i$

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8 Find the two square roots of $-8-6i$ in Cartesian form.

a. $\pm(1-3i)$

b. $\pm(3-i)$

c. $\pm(1+3i)$

d. $\pm(2-3i)$

9 Solve the quadratic equation $z^2-4z+13=0$ over $\mathbb{C}$.

a. $z=2\pm3i$

b. $z=4\pm6i$

c. $z=-2\pm3i$

d. $z=2\pm9i$

Section 4: Argand Diagram Geometry & Loci

10 Describe the locus of points in the complex plane satisfying $|z-3+2i|=5$.

a. A circle with center $(3,\ -2)$ and radius 5

b. A circle with center $(-3,2)$ and radius 5

c. A straight line perpendicular to $3-2i$

d. An ellipse with focal distance 5

11 What geometric locus is defined by $|z-1|=|z-3i|$?

a. The perpendicular bisector of the segment connecting $(1,0)$ and $(0,3)$

b. A circle passing through $(1,0)$ and $(0,3)$

c. A line segment connecting $(1,0)$ and $(0,3)$

d. A parabola opening upward

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12 Find the Cartesian equation of the locus $\text{Arg}(z-2)=\frac{\pi}{4}$.

a. The half-line $y=x-2$ for $x>2$

b. The full line $y=x-2$

c. The circle $(x-2)^2+y^2=1$

d. The half-line $y=-x+2$ for $x<2$

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